AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

T216A
Cisco Networking (CCCNA) part A

Final Examination
Spring Semester 2024/2025

Date: 11/June/2025

Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed:** 3h

Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part 1: Multiple choices. 1 mark per question. **Total: 5 Marks**

1. Which characteristic describes cut-through switching?
   a. Error-free fragments are forwarded, so switching occurs with lower latency.
   b. Buffering is used to support different Ethernet speeds.
   c. Only outgoing frames are checked for errors.
   d. Frames are forwarded without any error checking.
2. Which switching method ensures that the incoming frame is error-free before forwarding?
   a. fragment free
   b. cut-through
   c. FCS
   d. store-and-forward
3. A network contains multiple VLANs spanning multiple switches. What happens when a device in VLAN 20 sends a broadcast Ethernet frame?
   a. All devices in all VLANs see the frame.
   b. Only devices that are connected to the local switch see the frame.
   c. Only devices in VLAN 20 see the frame.
   d. Devices in VLAN 20 and the management VLAN see the frame.
4. Which message does an IPv4 host use to reply when it receives a DHCPOFFER message from a DHCP server?
   a. DHCPOFFER
   b. DHCPREQUEST
   c. DHCPDISCOVER
   d. DHCPACK
5. A network administrator has configured an EtherChannel between two switches that are connected via four trunk links. If the physical interface for one of the trunk links changes to a down state, what happens to the EtherChannel?
   a. The EtherChannel will remain functional.
   b. The EtherChannel will transition to a down state.
   c. Spanning Tree Protocol will recalculate the remaining trunk links.
   d. Spanning Tree Protocol will transition the failed physical interface into forwarding mode.

Part 2: Answer all the questions. 4 marks per question **Total: 16 Marks**

1. Answer the following questions:
   a) What type of message can be used to test the reachability of a host on an IP network? (2 marks)
   b) If a host is trying to communicate with a server located on a different LAN and the default gateway of this host is incorrect. (2 marks)
2. Answer the following questions:
   a) Explain what will happen when the UDP protocol at the destination host receives datagrams that are not in the correct order as they have been sent from the source host, and the reason behind it. (2 marks)
   b) What is the purpose of the POP protocol and on which device is it installed? (2 marks)
3. Answer the following questions:
   a) **Hardware threats** is one of the four classes of physical threats; briefly explain this type of threat. (2 marks)
   b) To protect network devices, it is important to use strong passwords. There are many guidelines to follow when creating a strong password; mention two: (2 marks)

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4. Answer the following questions:
   a) How can redundancy be accomplished? (2 marks)
   b) How does a switch build it MAC address table? (2 marks)

Part 3: Answer all the following questions **Total: 29 Marks**

1. Answer the following questions related to VLSM (12 marks)
   The following network topology was provided by the network administrator:
   - **LAN1:** has 4 PCs, 4 VoIP phones, and 2 network printers connected to 2 switches*.
   - **LAN2:** has 10 PCs, 10 VoIP phones, 2 servers, 5 network cameras, and 1 network printer connected to 4 switches*.
   - **LAN3:** can support up to 32 wireless devices (Laptops, smart devices) connected via 5 access points**.
   - **LAN4:** has 1 PCs, 1 VoIP phones, 5 servers, and 2 network cameras connected to 1 switch*.

   * Switch(es) are NOT accessible remotely (no IP address is needed).
   ** Access point(s) are accessible remotely (an IP address should be assigned for the access point).

   a) Determine the number of hosts for each of the 4 networks? (4 marks)
   (Show the steps you followed to obtain the number of hosts)
   b) Determine the prefix length for each network (4 marks)
   (Show the steps you followed to obtain the prefix.)
   c) Subnet the address block 180.20.9.0/24 as appropriate to provide an efficient use of the addresses but at the same time provide sufficient addresses for all groups and links. Present your answer in a table similar to table1. (4 marks)

Network name	Network address	Host range	Broadcast address	Prefix (or subnet mask)

Table 1

2. Answer the following questions (7 marks)
   a) The IT administrator is configuring a router with SSH. (3 marks)
   Complete the missing parts of the configuration:

```
hostname R1
__A__ test password cisco
__B__ domain-name AOU.com
crypto key __C__ RSA
The name for the keys will be: R1.AOU.com
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.
How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
ip ssh __D__ 2
line vty 0 4
transport __E__ SSH
__F__ local
```


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a) Write the commands needed to configure **Router on a stick interVLAN Routing** for VLAN number 40 interface G1/1. (4 marks)
To do this complete the following tasks:

- Create the sub-interface
- Set the encapsulation type to 802.1Q and assign VLAN number 40 to the sub-interface
- Assign the following IP address 180.20.9.1 and subnet mask 255.255.255.128 to the sub-interface
- Activate the interface

Note: five commands are needed to complete this part

3. Answer the following questions (10 marks)
   a) Compress the following IPv6 address: (2 marks)
   2001:0000:0000:0000:0000:0002:9000:08E0
   b) Decompress the following IPv6 address: (2 marks)
   2008:9::AB
   c) Answer the following questions related to STP (6 marks)

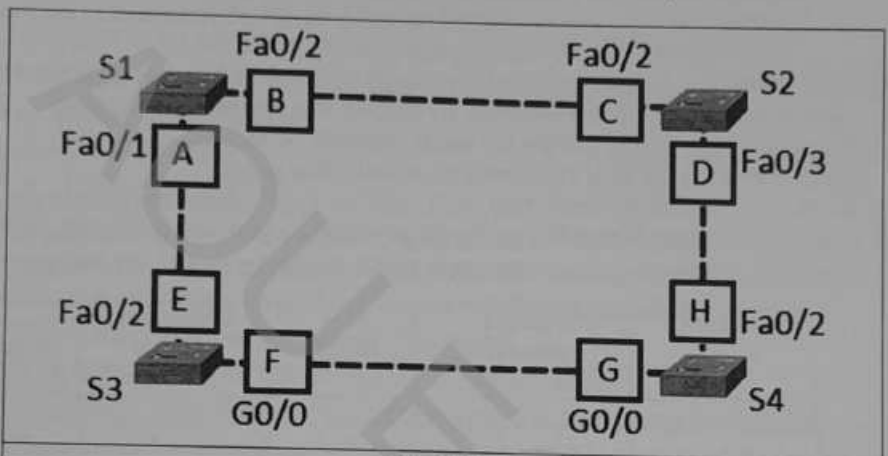


Figure 1

	Priority	MAC address
S1	32769	00:00:00:00:00:02
S2	28673	00:00:0A:00:00:03
S3	24577	AB:00:79:00:00:15
S4	36865	00:00:00:00:00:01

Table A

Default port costs		
Port	**Link speed**	**Cost (Revised IEEE specification)**
10 Gigabit Ethernet "10G"	10 Gb/s	2
Gigabit Ethernet "G"	1 Gb/s	4
Fast Ethernet "Fa"	100 Mb/s	19
Ethernet "E"	10 Mb/s	100

Table B

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c1) Identify which of the four switches figure 1 will become the Root Bridge by the STP protocol (Spanning Tree Protocol). Explain the reason (1 mark)

c2) From the answer provided in part b1 determines for each of the 4 switches (as shown in figure 1) the role each port will take. (5 marks)

The role a port can take are:

- Root port
- Designated port
- Alternate port

Note: take into consideration the interface cost (cf. table B).

End of questions

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